

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 8: Introduction to Trigonometry

Previous year board questions, topic-wise: 27 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Trigonometric ratios

1. If $\sin \theta = \frac{a}{b}$, write $\cos \theta$ in terms of a and b . (a) $\frac{b}{\sqrt{b^2 - a^2}}$ (b) $\frac{b}{a}$ (c) $\frac{\sqrt{b^2 - a^2}}{b}$ (d) $\frac{a}{\sqrt{b^2 - a^2}}$
CBSE 2026, 1 mark
2. If $\tan \theta = \frac{24}{7}$, find $\sin \theta + \cos \theta$. CBSE 2026, 2 marks
3. If $\cot \theta = \frac{7}{8}$, evaluate $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$. CBSE 2026, 2 marks
4. If $2 \tan A = 3$, find the value of $\frac{4 \sin A + 3 \cos A}{4 \sin A - 3 \cos A}$. (a) $\frac{7}{\sqrt{13}}$ (b) $\frac{1}{\sqrt{13}}$ (c) 3 (d) does not exist
CBSE 2023, 1 mark
5. If $\sin \alpha = \frac{1}{\sqrt{2}}$ and $\cot \beta = \sqrt{3}$, find $\operatorname{cosec} \alpha + \operatorname{cosec} \beta$. CBSE 2023, 2 marks
6. If $4 \tan \theta = 3$, evaluate $\frac{4 \sin \theta - \cos \theta + 1}{4 \sin \theta + \cos \theta - 1}$. CBSE 2018, 3 marks

2 Ratios of standard angles

7. If $\cos A = \frac{1}{2}$, find the value of $\sin^2 A + 2 \cos^2 A$. (a) $\frac{3}{2}$ (b) $\frac{5}{4}$ (c) -1 (d) $\frac{1}{2}$ CBSE 2026, 1 mark
8. Find the acute angle θ for which $7 + 4 \sin \theta = 9$. CBSE 2025, 1 mark
9. Given that $x \cos 60^\circ + y \cos 0^\circ + \sin 30^\circ - \cot 45^\circ = 5$, find the value of $x + 2y$. CBSE 2025, 2 marks
10. Evaluate $\frac{\tan^2 60^\circ}{\sin^2 60^\circ + \cos^2 30^\circ}$. CBSE 2025, 2 marks
11. Given $\cos(\alpha + \beta) = 0$, find $\cos\left(\frac{\alpha + \beta}{2}\right)$. (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$ (c) 0 (d) $\sqrt{2}$ CBSE 2024, 1 mark
12. Find the value of $2\sqrt{2} \cos 45^\circ \sin 30^\circ + 2\sqrt{3} \cos 30^\circ$. CBSE 2024, 2 marks
13. For $A = 60^\circ$ and $B = 30^\circ$, verify that $\sin(A + B) = \sin A \cos B + \cos A \sin B$. CBSE 2024, 2 marks
14. Evaluate $\frac{3}{4} \tan^2 30^\circ - \sec^2 45^\circ + \sin^2 60^\circ$. (a) -1 (b) $\frac{5}{6}$ (c) $-\frac{3}{2}$ (d) $\frac{1}{6}$ CBSE 2023, 1 mark

3 Trigonometric identities

15. Given $x = h + a \cos \theta$ and $y = k + b \sin \theta$, prove that $\left(\frac{x-h}{a}\right)^2 + \left(\frac{y-k}{b}\right)^2 = 1$. CBSE 2026, 3 marks

16. Prove that $\frac{\tan A}{1 + \sec A} - \frac{\tan A}{1 - \sec A} = 2 \operatorname{cosec} A$. CBSE 2026, 3 marks
17. Simplify $\tan^2 \theta - \left(\frac{1}{\cos \theta} \times \sec \theta \right)$. CBSE 2025, 1 mark
18. Prove that $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$. CBSE 2025, 3 marks
19. Prove that $\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} = \frac{2}{2 \sin^2 A - 1}$. CBSE 2025, 3 marks
20. Given $\sec \theta - \tan \theta = m$, find $\sec \theta + \tan \theta$. (a) $1 - \frac{1}{m}$ (b) $m^2 - 1$ (c) $\frac{1}{m}$ (d) $-m$ CBSE 2024, 1 mark
21. Prove the identity $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$. CBSE 2024, 3 marks
22. If $\sin \theta + \cos \theta = \sqrt{3}$, find the value of $\sin \theta \cos \theta$. CBSE 2023, 2 marks
23. Prove that $\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1} = \frac{1 + \sin \theta}{\cos \theta}$. CBSE 2023, 3 marks
24. Prove that $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta$. CBSE 2019, 3 marks
25. Prove that $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$. CBSE 2019, 3 marks
26. Prove that $\frac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = \frac{1}{\sec A - \tan A}$. CBSE 2019, 4 marks
27. Prove that $\frac{\sin A - 2 \sin^3 A}{2 \cos^3 A - \cos A} = \tan A$. CBSE 2018, 4 marks

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